Signed bias (days)

Median/cutoff process calibration with and without TabM residual correction · scoring dates

20/20 requested configurations complete · 2026-09-13 07:10:57 UTC

Process model
RAW · Calibration

	07	65	81	89
FL	+0.4	+0.9	+0.0	-0.1
FB	+2.2	+4.1	-4.1	+1.2
CL-FL-S	+0.7	+1.0	-0.8	+1.4
CL-FL-P	+0.9	+1.0	-0.2	+0.3
CB-FL-S	+0.9	+1.0	-0.8	+0.4
CB-FL-P	+1.0	+0.9	-0.8	-0.0
CL-FB-S	+2.1	-2.5	-3.1	+1.8
CL-FB-P	+0.7	-4.6	-6.4	+1.4
CB-FB-S	+2.0	-1.6	-2.9	+1.6
CB-FB-P	+3.5	-4.1	-5.9	+1.3

Process model
RAW · Test

	07	65	81	89
FL	+0.3	+0.5	-1.7	-1.8
FB	+2.2	+3.6	-6.1	-0.2
CL-FL-S	+0.4	+0.7	-1.2	-0.9
CL-FL-P	+0.7	+0.7	-1.8	-1.3
CB-FL-S	+0.6	+0.7	-1.6	-1.8
CB-FL-P	+0.9	+0.7	-2.3	-1.6
CL-FB-S	+2.0	-2.6	-4.8	+0.1
CL-FB-P	+0.6	-5.0	-8.4	-0.1
CB-FB-S	+1.9	-1.7	-4.5	-0.0
CB-FB-P	+3.4	-4.4	-7.9	-0.2

Process model
DEM · Calibration

	07	65	81	89
FL	+0.4	+0.6	-0.2	+0.2
FB	-0.1	+0.5	-0.6	+1.4
CL-FL-S	+0.9	+0.9	-0.9	+0.8
CL-FL-P	+1.0	+1.0	-0.2	+0.9
CB-FL-S	+0.7	+1.0	-0.5	+0.8
CB-FL-P	+1.0	+1.0	-0.2	+0.8
CL-FB-S	-0.0	+0.8	+0.2	+2.3
CL-FB-P	+1.0	+0.8	-0.3	+1.2
CB-FB-S	+0.8	+0.9	-0.4	+2.3
CB-FB-P	+1.0	+0.8	-0.3	+1.3

Process model
DEM · Test

	07	65	81	89
FL	+0.1	+0.1	-1.8	-2.2
FB	-0.4	-0.2	-2.2	-1.0
CL-FL-S	+1.0	+0.6	-1.9	-2.1
CL-FL-P	+0.6	+0.7	-1.6	-1.5
CB-FL-S	+0.4	+0.7	-1.5	-2.1
CB-FL-P	+0.7	+0.6	-1.6	-1.7
CL-FB-S	-0.0	+0.3	-1.1	-0.5
CL-FB-P	+0.5	+0.1	-1.9	-1.0
CB-FB-S	+0.9	+0.5	-1.7	-0.4
CB-FB-P	+0.5	+0.1	-1.9	-0.9

Process model + TabM residual
RAW · Calibration

	07	65	81	89
FL	-0.2	+0.5	-0.2	+0.2
FB	-0.2	+0.3	+1.3	+2.0
CL-FL-S	+1.0	+0.1	+1.2	+1.4
CL-FL-P	-0.0	-0.1	+0.5	-0.2
CB-FL-S	+0.3	+0.3	-0.2	+0.9
CB-FL-P	-0.1	+0.4	+0.3	-0.6
CL-FB-S	+1.0	+0.6	+0.6	+1.9
CL-FB-P	-0.7	+0.4	+0.3	+1.3
CB-FB-S	-0.3	-0.2	+0.0	+1.7
CB-FB-P	-0.9	+0.2	-1.1	+0.5

Process model + TabM residual
RAW · Test

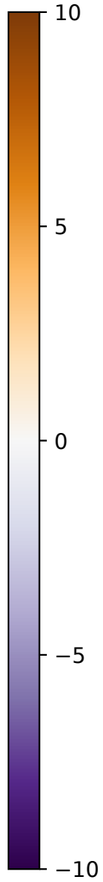
	07	65	81	89
FL	-0.6	+0.2	-1.7	-1.0
FB	-0.5	-0.2	-0.3	+0.2
CL-FL-S	+0.7	+0.1	+0.1	-0.7
CL-FL-P	-0.5	-0.3	-0.9	-1.6
CB-FL-S	-0.0	+0.1	-1.3	-1.5
CB-FL-P	-0.5	+0.1	-1.3	-1.9
CL-FB-S	+0.6	+0.2	-1.1	-0.3
CL-FB-P	-1.1	-0.1	-1.5	-0.5
CB-FB-S	-0.7	-0.5	-1.6	-0.0
CB-FB-P	-1.3	-0.2	-2.7	-1.2

Process model + TabM residual
DEM · Calibration

	07	65	81	89
FL	+0.4	+0.2	-0.1	+1.2
FB	+0.4	-0.1	+0.2	+1.0
CL-FL-S	+0.2	+0.1	+0.5	+0.9
CL-FL-P	+0.2	+0.2	+0.2	+0.3
CB-FL-S	-0.2	+0.0	+0.4	+1.5
CB-FL-P	+0.3	+0.5	+0.3	-0.4
CL-FB-S	-0.5	-0.0	+1.7	+1.3
CL-FB-P	-0.6	+1.0	+0.8	+1.3
CB-FB-S	+0.1	+0.6	+1.2	+1.7
CB-FB-P	-0.4	-0.1	+0.3	+1.2

Process model + TabM residual
DEM · Test

	07	65	81	89
FL	-0.0	-0.5	-1.5	-1.1
FB	-0.0	-0.9	-1.4	-1.6
CL-FL-S	-0.1	-0.4	-1.0	-1.8
CL-FL-P	-0.1	-0.4	-1.2	-1.9
CB-FL-S	-0.5	-0.5	-1.1	-1.5
CB-FL-P	-0.2	+0.0	-1.0	-2.7
CL-FB-S	-0.7	-0.6	+0.3	-1.9
CL-FB-P	-1.3	+0.2	-0.8	-1.1
CB-FB-S	-0.2	-0.0	-0.2	-1.2
CB-FB-P	-1.1	-0.8	-1.3	-1.2



BBCH 07 budburst · 65 flowering · 81 veraison · 89 maturity/harvest. Grey —: pending; ×: failed; ·: unrequested; n/a: unavailable in a completed unit.

MAE and bias average within sites, then weight sites equally. Bias = predicted − observed; negative is early.

TabM residual target = observed raw date − process-model raw date; features include weather through process-predicted dates.

Final hybrid dates undergo chronological correction before BBCH81/89 scoring applies the 4 November cap.

An unattained process prediction receives no ML residual; BBCH81/89 can still receive 4 November for scoring.

Calibration panels show apparent training performance, not OOF estimates. Test sites are held out. No confidence intervals.